

ENVARIABELSÄNALYS DEL 2

SEMINARIUM: RATIONELLA UTTRYCK

(SE TEORI: S.3-S.5)

$$\int \frac{f(x)}{g(x)} dx$$

• $f(x)$ & $g(x)$ är polynom

① Om $\text{grad } f \geq \text{grad } g$ utför polynom division först.

$$\text{Då fås: } \frac{f(x)}{g(x)} = h(x) + \frac{\text{rest}}{g(x)}$$

② Om $\text{grad } f < \text{grad } g$ utför partialbräksuppdelning.

a) Faktorisera först nämnaren.

b) Gör ansats.

c) Bestäm konstanter (gärna med handräkning).

OBS:

$$\int \frac{g'(x)}{g(x)} dx = \ln|g(x)| + c$$

"Inom partiell bräksuppdelning"

OBS:

I svar om $\ln|x+1| + c$ är

Svaret alltid $F(x) = \ln(x+1) + c$

Men $\ln|\cos x|$ är $F(x) = \ln|\cos x| + c$

Ex:

$$\int \left(\frac{2x-1}{x^2+2x+3} \right) dx = \int \left(\frac{2x-1}{(x+1)^2-2^2} \right) dx =$$

$$= \int \left(\frac{2x-1}{(x+3)(x-1)} \right) dx =$$

$$\frac{2x-1}{(x+3)(x-1)} \stackrel{\text{Partialbråk / uppdelning}}{=} \frac{A}{(x-1)} + \frac{B}{(x+3)}$$

"hand påläggning"

$$\frac{2x-1}{(\cancel{x})(x+3)} = \frac{\cancel{A}}{\cancel{x}} + \frac{B}{(x+3)} \Rightarrow A = \frac{1}{4}$$

$$\frac{2x-1}{(x-1)(\cancel{x})} = \frac{A}{(x-1)} + \frac{\cancel{B}}{\cancel{x}} \Rightarrow B = \frac{-7}{-4} = \frac{7}{4}$$

"utan hand påläggning"

$$\frac{2x-1}{(x-1)(x+3)} = \frac{A}{(x-1)} + \frac{B}{(x+3)} =$$

$$\frac{2x-1}{(x-1)(x+3)} = \frac{A(x+3)}{(x+3)(x-1)} + \frac{B(x-1)}{(x+3)(x-1)}$$

$$\frac{2x-1}{(x-1)(x+3)} = \frac{A(x+3) + B(x-1)}{(x+3)(x-1)}$$

$$2x-1 = Ax + 3A + Bx - B$$

$$2x-1 = x(A+B) + (3A-B)$$

$$\begin{cases} 2 = A+B \\ -1 = 3A-B \end{cases} \quad | \cdot (-1) + \cdot (-2) | \quad A = \frac{1}{4}$$

$$\text{lös ger } B = \frac{7}{4}$$

Då fås

$$\int \left(\frac{1/4}{x-1} + \frac{7/4}{x+3} \right) dx = \frac{1}{4} \int \frac{1}{x-1} dx + \frac{7}{4} \int \frac{1}{x+3}$$

$$= \frac{1}{4} \ln|x-1| + \frac{7}{4} \ln|x+3| + C$$

SVAR: $F(x) =$

Ex: Korrekta ansatser

$$\int \frac{5x+7}{x(x-1)^3(x^2+1)} dx = [\text{Parbital braks uppdelning}]$$

$$= \frac{A}{x} + \frac{B}{(x-1)} + \underbrace{\frac{C}{(x-1)^2} + \frac{D}{(x-1)^3}}_{\text{Multiplar av } \frac{1}{x-1} \text{ så } C \text{ o } D.} + \frac{Ex+F}{x^2+1}$$

Ex:

$$\frac{1}{x^3-x} = \frac{1}{x(x^2-1)} = \frac{1}{x(x+1)(x-1)} = \frac{A}{x} + \frac{B}{(x-1)} + \frac{C}{(x+1)}$$

$$\frac{1}{x(x+1)(x-1)} = \frac{A}{x} + \frac{B}{(x-1)} + \frac{C}{(x+1)}$$

$$A = \frac{1}{0(0+1)(0-1)} = \frac{1}{1(-1)} = -1$$

$$\frac{1}{x(x+1)(x-1)} = \frac{A}{x} + \frac{B}{(x-1)} + \frac{C}{(x+1)}$$

$$B = \frac{1}{1(1+1)(1-1)} = \frac{1}{2}$$

$$\frac{1}{x(x-1)(x+1)} = \frac{A}{x} + \frac{B}{(x-1)} + \frac{C}{(x+1)}$$

$$C = \frac{1}{(-1)(-1+1)(-1-1)} = -\frac{1}{2}$$

OBS:

Kompensera alltid för ihre derivatan, men bara om den är konstant (annars variabelbyte rekommenderas).

$$\text{tex: } \int \frac{2}{1-x} dx = 2 \int \frac{1}{1-x} dx = -2 \int \frac{-1}{1-x} dx$$

$$= -2 \ln|1-x| + C$$

Här kompenserar vi med -1!

Ex: $\int \left(\frac{x+1}{x^2+2x+5} \right) dx = \left[\begin{array}{l} x^2+2x+5 = \\ = (x+1) + 4 \\ \text{kan ej faktorisera!} \\ \text{vertex} \\ (-1, 4) \\ \text{Skär ej x-axeln} \end{array} \right]$

$$= \int \left(\frac{1}{2} \cdot \frac{2x+2}{x^2+2x+5} \right) dx$$

$$= \frac{1}{2} \int \left(\frac{2x+2}{x^2+2x+5} \right) dx$$

$$= \left[\int \frac{g(x)}{g(x)} dx = \ln|xt| + C \right]$$

$$= \frac{1}{2} \ln|x^2+2x+5| + C$$

$$F(x) = \frac{1}{2} \ln(x^2+2x+5) + C$$

Den är positiv så byter
sin parentes ist för absolut

Ex: $\int \frac{x+2}{x^2+2x+5} dx = \left[\begin{array}{l} \text{Närman} \\ \text{kan ej} \\ \text{faktorisera} \end{array} \right] = \frac{1}{2} \int \left(\frac{2x+4}{x^2+2x+5} \right) dx$

$$= \frac{1}{2} \int \left(\frac{(2x+2)+2}{x^2+2x+5} \right) dx = \frac{1}{2} \int \frac{2x+2}{x^2+2x+5} dx + \frac{1}{2} \int \frac{2}{x^2+2x+5} dx$$

$$= \frac{1}{2} \ln|x^2+2x+5| + \int \frac{1}{x^2+2x+5} dx$$

$$= \frac{1}{2} \ln|x^2+2x+5| + \int \left(\frac{1}{(x+1)^2+4} \right) dx$$

$$= \left[\begin{array}{l} x+1 = t \\ dx = dt \end{array} \right] = \frac{1}{2} \ln|x^2+2x+5| + \int \frac{1}{t^2+4} dt$$

$$= \frac{1}{2} \ln|x^2+2x+5| + \int \frac{1}{4 \left(\left(\frac{t}{2} \right)^2 + 1 \right)} dt$$

$$= \frac{1}{2} \ln|x^2+2x+5| + \int \left(\frac{1}{4} \cdot \frac{1}{\left(\frac{t}{2} \right)^2 + 1} \right) dt$$

$$= \left[\begin{array}{l} \frac{t}{2} = u \\ dt = 2du \end{array} \right] = \frac{1}{2} \ln|x^2+2x+5| + \frac{1}{4} \int \frac{1}{u^2+1} 2du$$

$$= \frac{1}{2} \ln |x^2 + 2x + 5| + \frac{1}{2} \int \frac{1}{u^2 + 1} du$$

$$= \frac{1}{2} \ln |x^2 + 2x + 5| + \frac{1}{2} \arctan(u) + C$$

$$[u = \frac{t}{2}]$$

$$\frac{1}{2} \ln |x^2 + 2x + 5| + \frac{1}{2} \arctan\left(\frac{t}{2}\right) + C$$

$$[t = x+1]$$

$$\frac{1}{2} \ln |x^2 + 2x + 5| + \frac{1}{2} \arctan\left(\frac{x+1}{2}\right) + C$$

SVAR:

$$F(x) = \frac{1}{2} \ln(x^2 + 2x + 5) + \frac{1}{2} \arctan\left(\frac{x+1}{2}\right) + C$$

ELSA EK:

$$\int e^x \sin(x) dx = \begin{bmatrix} u = \sin(x) & u' = \cos(x) \\ v = e^x & v' = e^x \end{bmatrix}$$

$$= \sin(x) e^x - \int e^x \cos(x) dx$$

$$= \begin{bmatrix} u = \cos(x) & u' = -\sin(x) \\ v = e^x & v' = e^x \end{bmatrix}$$

$$= e^x \sin(x) - (e^x \cos(x) - \int (-\sin(x) e^x) dx)$$

$$e^x \sin(x) - e^x \cos(x) + \int (-\sin(x) e^x) dx$$

$$= \frac{e^x \sin(x) - e^x \cos(x)}{e^x (\sin(x) - \cos(x))} = \int (\sin(x) e^x) dx$$

Satt nu lika med vad vi hade från början!

$$\int e^x \sin(x) dx = e^x (\sin(x) - \cos(x)) - \int \sin(x) e^x dx$$

$$2 \int e^x \sin(x) dx = e^x (\sin(x) - \cos(x))$$

$$\int e^x \sin(x) dx = \frac{1}{2} e^x (\sin(x) - \cos(x)) + C$$

$$F(x) =$$